

s/Math + AI/ Reasoning in LLMs

PRECOG Recruitment Task

Aryaman Bahl

IIIT Hyderabad

December 17, 2025

Research Question

Can we evaluate LLM reasoning capabilities using controlled symbolic reasoning tasks?

Four Tasks Completed:

- **Task 1:** Dataset Generation (100 SED puzzles)
- **Task 2:** LLM Evaluation (3 models, 5 prompts)
- **Task 3:** Evaluation Metrics (4 metrics, identifiability analysis)
- **Task 4 (Bonus):** Human vs Machine (cognitive asymmetry)

Key Finding: Surface metrics (validity, progress) fail to identify true reasoning competence—even in controlled domains.

What Was Done & Not Done

Completed:

- All 4 tasks (including bonus)
- 100-puzzle dataset generation
- Comprehensive LLM evaluation
- Multi-metric analysis
- Identifiability theory
- Human-machine comparison
- Reading Task
- Full codebase + notebooks

Not Done:

- Expert-level puzzles ($D \geq 75$)
- Additional model families (GPT, Claude)
- Human evaluation study

Why:

- Controlled domain analysis
- Theoretical contributions prioritized

Dataset Generation:

- Two-phase pipeline
- Constructive + BFS verification
- Difficulty stratification
- Quality filtering

LLM Evaluation:

- API-based inference
- Solution parsing & validation
- Step-by-step execution
- Error categorization

Metric Analysis:

- Correlation analysis
- Edge case decomposition
- Identifiability theory
- Disagreement quantification

Human Comparison:

- Generator-stratified analysis
- Structural difficulty heuristics
- Cognitive framework

Task 1: SED Puzzle Dataset

Problem Definition:

- Initial string: "ABCDEF"
- Transformation rules: "ABC" $\rightarrow$ ""
- Goal: Reach empty string ""

Dataset:

- 100 puzzles
- 8 generator types
- 3 difficulty levels
- BFS-verified solutions

Generator Types:

- backward_3/5/7: Sequential
- concat_2: Pattern matching
- sort_3/4: Algorithmic
- palin_3: Symmetry
- multiphase: Multi-phase

Difficulty Distribution:

- Easy: $D < 25$
- Medium: $25 \leq D < 50$
- Hard: $50 \leq D < 75$

Task 1: Dataset Characteristics

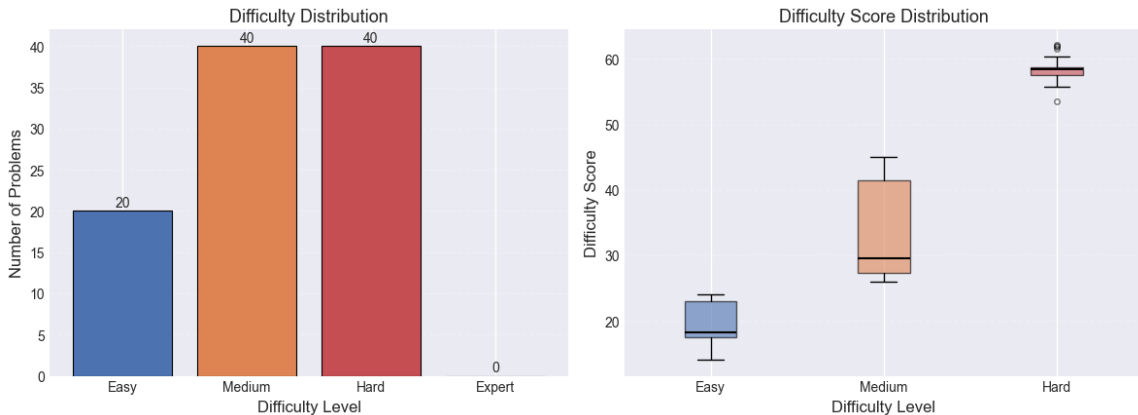


Figure: Distribution of SED puzzles by difficulty level (left) and corresponding difficulty score distributions (right).

Task 1: Dataset Characteristics

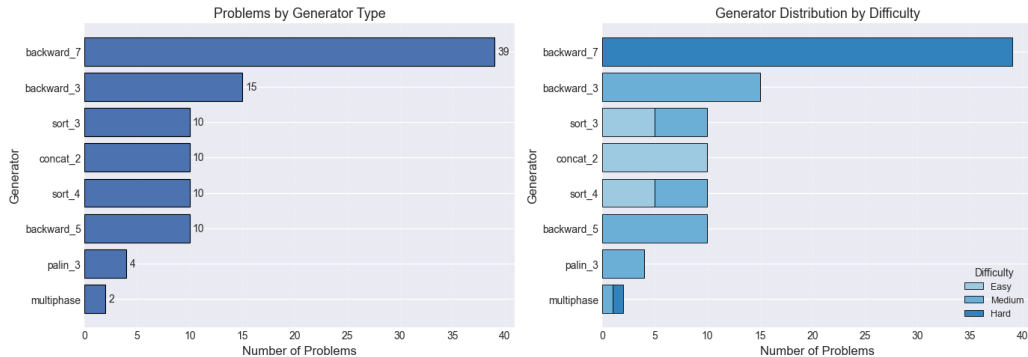


Figure: Puzzle distribution by generator type and difficulty

Task 2: LLM Evaluation Setup

Models:

- LLaMA 3.3 70B
- LLaMA 4 Scout 17B
- LLaMA 3.1 8B

Prompting Strategies:

- Zero-Shot
- Few-Shot (5 examples)
- Chain-of-Thought
- Self-Verify
- Role-Classify

Evaluation:

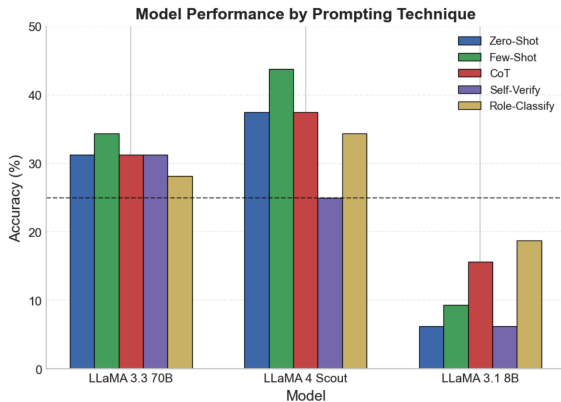
- 32 test puzzles
- 480 total evaluations
- Binary accuracy metric

Best Performance:

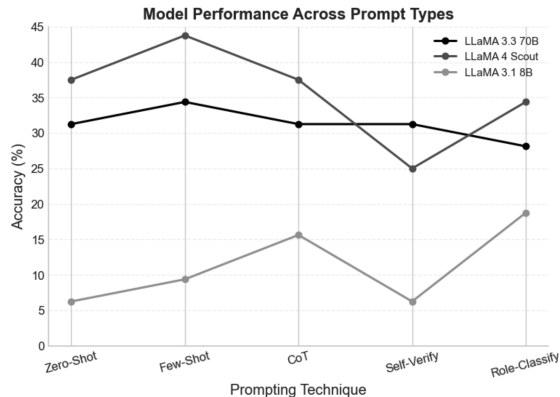
- LLaMA 4 Scout + Few-Shot
- **43.8% accuracy**

Overall: 26% accuracy

Task 2 Prompting Effects Across Models



Accuracy trends across prompting techniques for each model.



Model-wise accuracy comparison showing limited and inconsistent gains from advanced prompting.

Task 2: Difficulty Dominates Performance

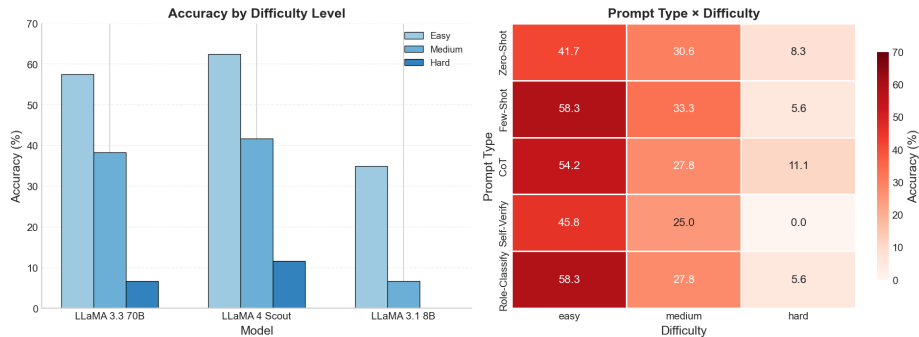


Figure: Accuracy drops sharply with difficulty: Easy (51.7%) → Hard (6.1%)

Key Insight: Difficulty is the dominant factor, not model size or prompt complexity.

Key Findings

- **Low overall accuracy:** 26% accuracy over 480 evaluations on a 32-puzzle test set.
- **Scale is not enough:** LLaMA-4 Scout (17B) outperforms LLaMA-3.3-70B.
- **Limited gains from prompting:** Few-shot helps slightly; complex prompts add cost without consistent gains.
- **Difficulty dominates:** Accuracy drops from 51.7% (easy) to 6.1% (hard).
- **Errors are systematic:** Failures stem from invalid reasoning, not formatting or execution errors.

Prompt engineering alone does not resolve reasoning failures.

Task 3: Evaluation Metrics

Four Metrics Defined

- 1 **Binary Accuracy:** $A_{\text{bin}} = 1[s_f = \varepsilon]$
- 2 **Progress Score:** $P = \frac{|s_0| - |s_f|}{|s_0|}$
- 3 **Valid Steps Ratio:** $V = \frac{n_{\text{valid}}}{n_{\text{total}}}$
- 4 **Composite Score:** $C = 0.5A + 0.3P + 0.2V$

Findings: Validity is **non-identifying**— $\rho(V, A_{\text{bin}}) = 0.111$

- High validity does not imply correctness
- 41.9% of traces with $V \geq 0.8$ are still incorrect

Task 3: Composite Score Design

Composite Score Formula

$$C = 0.5 \cdot A_{\text{bin}} + 0.3 \cdot P + 0.2 \cdot V$$

Weight Justification:

- **Correctness (50%):** Primary goal is solving the puzzle
- **Progress (30%):** Partial credit for near-misses
- **Validity (20%):** Rule compliance is necessary but not sufficient

Performance:

- Correlation with binary accuracy: $\rho(C, A_{\text{bin}}) = 0.963$
- Provides continuous gradient for optimization
- Better discriminates between models than binary accuracy alone
- Average composite score (0.576) > binary accuracy (0.508), reflecting partial credit

Task 3: Metric Correlations

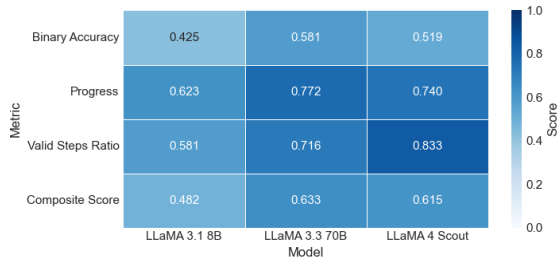
	A_{bin}	P	V	C
A_{bin}	1.000	0.753	0.111	0.963
P	0.753	1.000	0.286	0.861
V	0.111	0.286	1.000	0.330
C	0.963	0.861	0.330	1.000

Key Observation: Validity (V) is nearly **uncorrelated** with correctness (A_{bin}): $\rho = 0.111 \approx 0$.

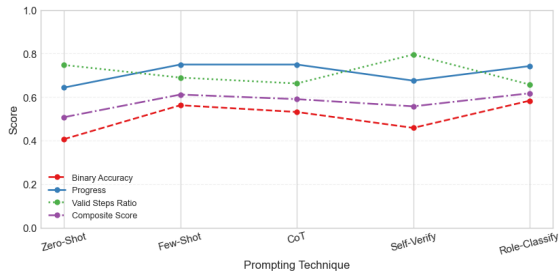
Implication: High syntactic validity does not imply semantic correctness. Models can achieve perfect validity ($V = 1.0$) while failing to solve the task.

Progress-Correctness Correlation: $\rho(P, A_{\text{bin}}) = 0.753$ indicates progress is moderately predictive, but substantial variance remains unexplained.

Task 3: Metric Correlation Analysis



Metric-accuracy correlation varies across model scales; validity aligns weakly with correctness.



Prompting changes surface metrics but not accuracy correlation.

Key Findings

- **Validity is non-identifying:** High step validity (including self-verification) correlates weakly with correctness.
- **Progress $\neq$ correctness:** Progress increases with scale ($0.623 \rightarrow 0.772$), while binary accuracy plateaus.
- **Few-shot is most effective:** Best binary accuracy (0.562) and composite score (0.611).

Task 3: Identifiability Analysis

Disagreement Mass

Probability that metrics indicate success while solution is incorrect:

$$D = \mathbb{P}(M_{\text{high}} \wedge S_{\text{fail}})$$

where $M_{\text{high}} = (P > 0.5) \vee (V > 0.8)$ and $S_{\text{fail}} = (s_{\text{final}} \neq \varepsilon)$

Empirical Results:

- **Overall:** $D = 0.277$ (133/480 cases)
- **Easy:** $D = 0.025$ (2.5% false positives)
- **Medium:** $D = 0.167$ (16.7% false positives)
- **Hard:** $D = 0.556$ (55.6% false positives)

Conclusion: Identifiability degrades **monotonically** with difficulty. On hard puzzles, metrics mislead evaluation in more than half of all cases.

Task 3: Disagreement by Difficulty

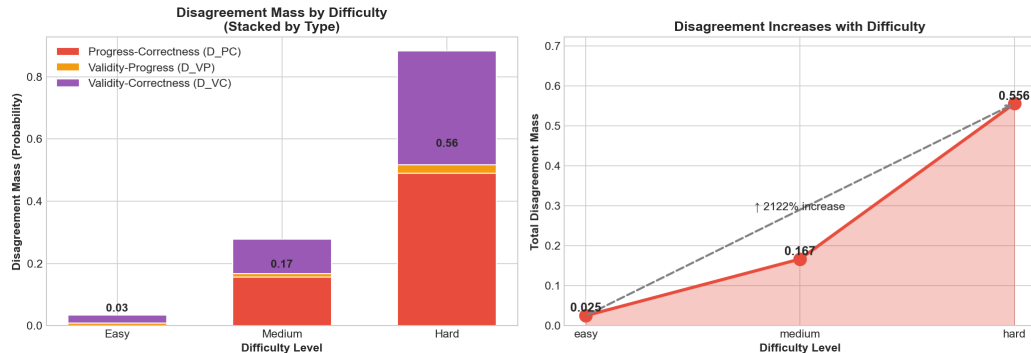
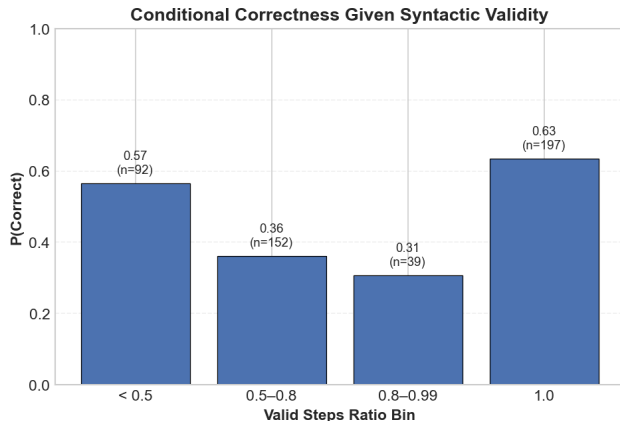


Figure: Disagreement mass increases monotonically with difficulty. The sharp increase from easy (2.5%) to hard (55.6%) demonstrates that surface metrics become unreliable proxies for reasoning competence as task complexity grows.

Task 3: High Validity Steps do not converge to Correct Solutions.



Key Findings

- **High validity $\neq$ correctness** 41.9% of traces with $V \geq 0.8$ are incorrect.
- **Near-perfect validity still fails** Even with $V \geq 0.99$, 36.5% of solutions are wrong.
- **Core implication** Syntactic rule compliance is *necessary but not sufficient* for semantic correctness.

This gap widens with task difficulty (disagreement mass analysis).

Task 3: Edge Cases: When Metrics Disagree

Edge Case	Count	Avg Progress	Avg Validity	Avg Composite
Correct Optimal	125	1.000	0.950	0.980
Correct Longer	119	1.000	0.650	0.820
Almost There	75	0.940	0.720	0.680
All Valid Wrong	72	0.580	1.000	0.350
Good Progress	61	0.850	0.600	0.550
Minimal Progress	100	0.250	0.400	0.200
Mostly Invalid	37	0.150	0.200	0.100

Key Insight: The "All Valid Wrong" category (72 cases) achieves **perfect validity** ($V = 1.0$) with **zero correctness** ($A_{\text{bin}} = 0$), demonstrating that syntactic compliance can occur without semantic understanding.

Task 3: Metric Selection by Use Case

Use Case	Recommended Metric	Rationale
Model Deployment	Binary Accuracy	Clear task completion signal
Model Ranking	Composite Score	Balances correctness and progress
Failure Analysis	Progress + Validity	Separates semantic vs syntactic failure
Training Feedback	Composite Score	Smooth, continuous gradient
Quick Benchmarking	Binary Accuracy	Simple, interpretable
Reasoning Quality	Progress + Qualitative	Reveals partial success patterns

Key Takeaway: No single metric is universally optimal. Metric choice must align with evaluation intent. Binary accuracy is appropriate for headline results, but deeper analysis requires graded metrics that capture partial progress and validity.

Task 4: Human vs Machine

Research Question

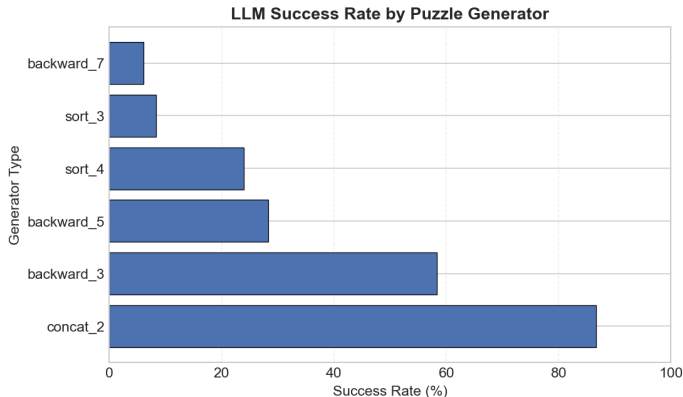
Do humans and LLMs exhibit systematically different difficulty profiles?

Three Empirical Categories:

- **Easy Human, Hard LLM:** Sorting puzzles (8–24% LLM vs. $\sim 100\%$ human)
- **Easy for Both:** Concatenation puzzles ($\sim 87\%$ LLM)
- **Hard for Both:** Long-chain puzzles ($\sim 6\%$ LLM)

Key Insight: Sorting puzzles expose an **84% human–LLM gap**: humans recognize algorithms, LLMs apply isolated rules.

Task 4: Success by Generator Type



Observation

- Performance varies sharply by generator.
- Sorting generators induce systematic failure.
- Reveals **cognitive asymmetries**, not noise.

Key Findings

- **Validity is non-identifying:** High step validity correlates weakly with correctness ($\rho = 0.111$).
- **Difficulty dominates performance:** Accuracy drops from 51.7% (easy) to 6.1% (hard).
- **Sorting reveals algorithmic blindness:** Trivially easy for humans, catastrophically hard for LLMs.
- **Metric reliability collapses on hard tasks:** Disagreement mass rises sharply with difficulty.

Evaluation

- Don't trust aggregate accuracy
- Stratify by reasoning type
- Include easy-human / hard-LLM cases

Model Design

- Iteration is the key gap
- Algorithm recognition matters

Benchmark Design

- Include algorithmic puzzles
- Avoid concat-only benchmarks
- Calibrate difficulty with humans

Theory

- Metric evaluation is an identifiability problem

Summary: Key Results

Metric	Value	Interpretation
Overall Accuracy	26.0%	Genuine difficulty
Best Model+Prompt	43.8%	LLaMA 4 Scout + Few-Shot
Easy Puzzles	51.7%	Pattern matching works
Hard Puzzles	6.1%	Planning fails
Validity-Correctness Correlation	0.111	Non-identifying
Progress-Correctness Correlation	0.753	Partially identifying
Disagreement Mass (Overall)	0.277	133/480 cases
Disagreement (Hard)	0.556	55.6% of hard cases
Sorting Puzzle Gap	84%	Human vs LLM

Main Thesis

Surface metrics fail to identify true reasoning competence, even in controlled symbolic domains.

Four Key Results:

- 1 **Validity is non-identifying** — $\rho(V, A) = 0.111$
- 2 **Difficulty dominates performance** — $51.7\% \rightarrow 6.1\%$
- 3 **Sorting puzzles expose algorithmic blindness** — 84% gap
- 4 **Disagreement increases with difficulty** — $D_{\text{hard}} = 0.556$

Takeaway: LLMs are not failing at reasoning in general—they are failing at recognizing and executing iterative algorithms.

Dataset:

- 100 SED puzzles
- 8 generator types
- 3 difficulty levels
- BFS-verified solutions

Evaluation:

- 3 LLaMA models
- 5 prompting strategies
- 480 total evaluations

Analysis:

- 4 evaluation metrics
- Correlation analysis
- Identifiability theory
- Human comparison

Codebase:

- GitHub repository
- Jupyter notebooks
- Reproducible pipeline

Limitations

- **Limited scale:** Small test set ($N = 32$) makes model ranking (e.g., Scout > 70B) inconclusive.
- **Heuristic metrics:** Composite score depends on chosen weights.
- **Validity gap:** High step validity often hides semantic failure, especially on hard tasks.

Next Steps

- **Scale evaluation:** Expand to $N \geq 200$ puzzles.
- **Human calibration:** Quantify cognitive gaps on sorting tasks.

Key Insight

Closing the validity gap likely requires **neuro-symbolic designs**: LLMs for strategy, symbolic engines for exact execution (cf. AlphaGeometry).

- Trinh, T.H., Wu, Y., Le, Q.V., He, H., & Luong, T. (2024). **Solving olympiad geometry without human demonstrations**. *Nature*, 625, 476–482. <https://doi.org/10.1038/s41586-023-06747-5>
- Wei, J., Wang, X., Schuurmans, D., et al. (2022). **Chain-of-Thought Prompting Elicits Reasoning in Large Language Models**. *NeurIPS 2022*. <https://arxiv.org/abs/2201.11903>
- Brown, T.B., Mann, B., Ryder, N., et al. (2020). **Language Models are Few-Shot Learners**. *NeurIPS 2020*. <https://arxiv.org/abs/2005.14165>
- Xia, W., Qin, L., & Radev, D. (2024). **Evaluating Mathematical Reasoning Beyond Accuracy: Validity and Redundancy in Step-by-Step Traces**. *arXiv preprint*. <https://arxiv.org/html/2404.05692v1>
- Goodhart, C. A. E. (1975). **Problems of Monetary Management: The U.K. Experience**. In *Papers in Monetary Economics*, Reserve Bank of Australia. (Origin of *Goodhart's Law*)
- Aurelius, M. (c. 167 AD). **Meditations**. Trans. G. Hays (2002). Modern Library.

GitHub Repository

<https://github.com/hanara2112/sed-reasoning>

Thank You

Questions?